

**COMPARATIVE ASSESSMENT OF RESTING INTRAOCULAR PRESSURE IN
REGULAR GYM-GOERS AND NON-GYM-GOERS IN ILORIN METROPOLIS,
KWARA STATE.**

BY

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**BEING A PROJECT SUBMITTED TO THE DEPARTMENT OF OPTOMETRY AND
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DOCTOR OF OPTOMETRY (O.D.) DEGREE**

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CERTIFICATION

This is to certify that the research project titled “COMPARATIVE ASSESSMENT OF RESTING INTRAOCULAR PRESSURE IN REGULAR GYM-GOERS AND NON-GYM-GOERS IN ILORIN METROPOLIS, KWARA STATE” was carried out by OFIKWU MATTHEW IDUH (19/55DZ038) in the Department of Optometry and Vision Science, Faculty of Life Sciences, University of Ilorin, Nigeria, under my supervision, in partial fulfillment of the requirements for the award of the Doctor of Optometry (O.D.) degree during the 2025/2026 academic session.

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APPROVAL PAGE

This project has been read and approved as having met part of the requirement of the Department of Optometry & Vision Science, University of Ilorin, Nigeria, for the award of a Doctor of Optometry (OD) degree.

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DEDICATION

This project is dedicated, first and foremost, to God — the giver of life and every good thing — whose grace has been the foundation of my journey and the source of strength that carried me through every challenge. To my parents and siblings, whose unwavering love, encouragement, and sacrifices made this work possible, this dedication is equally yours.

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ABSTRACT

This study compared resting intraocular pressure (IOP) between regular gym-goers and non-regular gym-goers in Ilorin metropolis, Kwara State, Nigeria. A comparative cross-sectional design was employed, with 120 adult participants (60 per group) recruited using a multistage sampling technique. IOP was measured using a calibrated tonometer, and data were analysed using descriptive statistics, independent-samples t-test, effect size estimation, and bivariate correlation in SPSS version 26.0. Results showed that gym-goers had a significantly lower mean resting IOP (14.82 ± 2.30 mmHg) than non-gym-goers (16.73 ± 3.22 mmHg; $t(106.885) = -3.741$, $p < .001$; Cohen's $d = 0.68$). No significant association was found between exercise frequency and IOP within the gym-goer subgroup ($r = .139$, $p = .288$). Regular gym participation is associated with lower resting IOP, with implications for glaucoma prevention and lifestyle counselling in optometric practice.

Keywords: *Intraocular pressure, gym-goers, physical exercise, glaucoma risk, Ilorin, Nigeria*

CHAPTER ONE

1.0 Introduction

Intraocular pressure (IOP) is a fundamental physiological parameter that plays a critical role in maintaining the structural integrity and optical function of the eye. It represents the pressure exerted by intraocular fluids against the walls of the globe and is primarily determined by the balance between aqueous humour production by the ciliary body and its drainage through the trabecular meshwork and uveoscleral pathways (Kanski & Bowling, 2016). The maintenance of this balance is essential for preserving normal ocular function, and any disruption may result in pathological changes that compromise vision.

Elevated intraocular pressure is widely recognized as the most significant modifiable risk factor for glaucoma, a chronic optic neuropathy characterized by progressive degeneration of retinal ganglion cells and corresponding visual field loss (Weinreb *et al.*, 2014). Glaucoma is one of the leading causes of irreversible blindness worldwide, and its burden continues to increase due to population growth and aging (Tham *et al.*, 2025). The condition is particularly concerning because it often progresses silently, with patients remaining asymptomatic until advanced stages, at which point substantial and irreversible vision loss may have already occurred.

The burden of glaucoma is disproportionately higher in sub-Saharan Africa compared to other regions of the world. Epidemiological studies have shown that individuals of African descent are at greater risk of developing glaucoma and are more likely to experience severe disease progression and poorer visual outcomes (Asiamah *et al.*, 2025). In Nigeria, population-based surveys indicate that glaucoma is a leading cause of blindness among adults, with a significant proportion of

affected individuals unaware of their condition due to late presentation and limited access to eye care services (Kyari *et al.*, 2015). This highlights the importance of early detection and the identification of modifiable factors that may influence intraocular pressure.

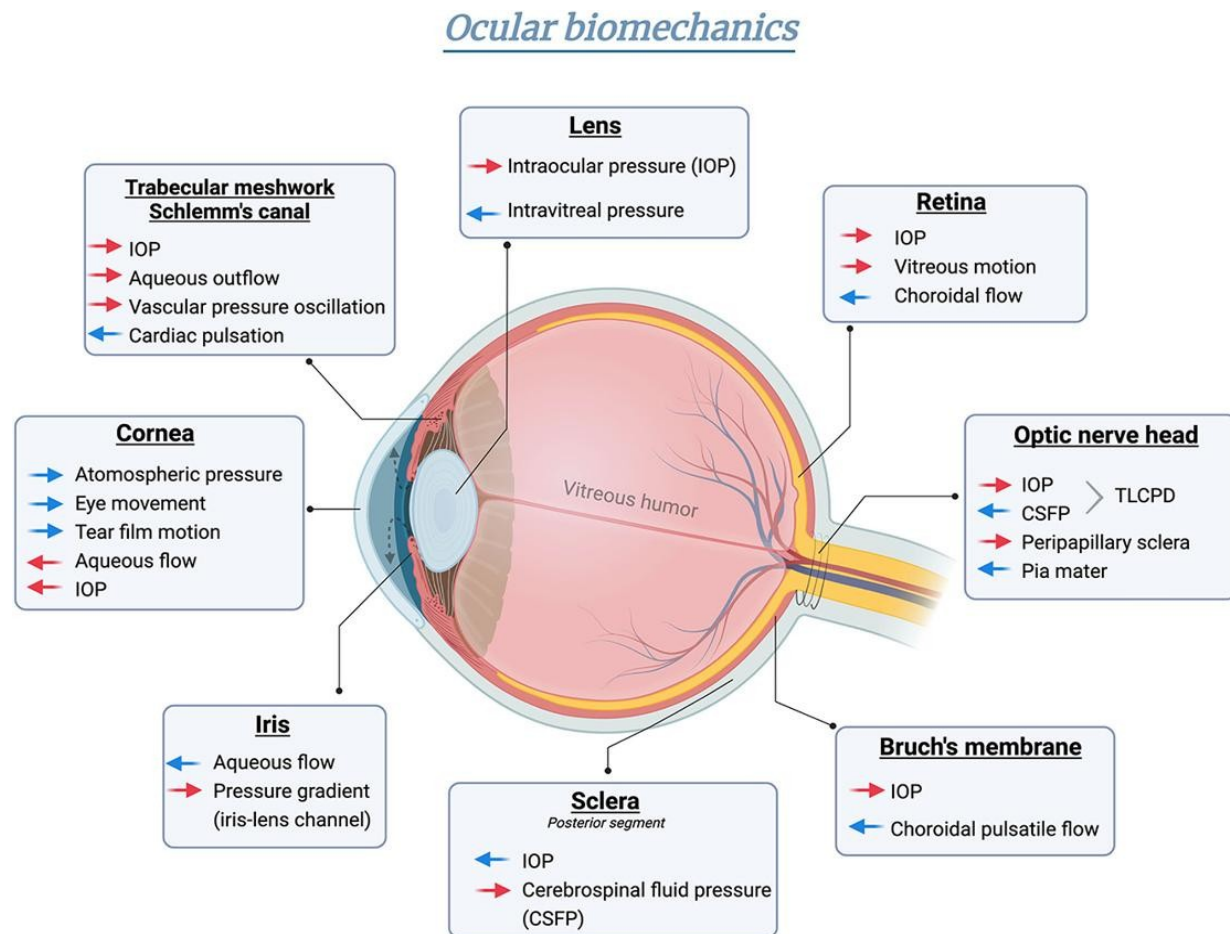


Figure 1: Ocular biomechanics showing the influence of intraocular pressure on ocular structures (Adapted from Downs *et al.* (2011) and Burgoyne (2011)).

In recent years, physical exercise has gained increasing attention as a potential lifestyle factor capable of influencing intraocular pressure. Exercise induces a wide range of physiological changes, including alterations in cardiovascular function, autonomic nervous system activity, metabolic processes, and vascular dynamics. These systemic changes may, in turn, influence ocular

physiology, particularly aqueous humour dynamics and episcleral venous pressure, which are key determinants of intraocular pressure (Harris *et al.*, 2008).

Aerobic exercise, such as running and cycling, has been associated with transient reductions in intraocular pressure, possibly due to improved ocular blood flow, reduced episcleral venous pressure, and enhanced autonomic regulation (Qureshi *et al.*, 1996; Vera *et al.*, 2017). In contrast, resistance exercise, particularly when associated with breath-holding and Valsalva manoeuvres, may lead to short-term increases in intraocular pressure due to elevated intrathoracic pressure (Vieira *et al.*, 2006). While these acute effects are relatively well documented, the long-term impact of habitual exercise on resting intraocular pressure remains less clearly understood.

Resting intraocular pressure, measured under baseline physiological conditions, is more clinically relevant than transient post-exercise changes, as it reflects the steady-state regulation of ocular pressure. However, most existing studies have focused on immediate post-exercise measurements rather than resting values, limiting their applicability to clinical practice and long-term glaucoma risk assessment (Vera *et al.*, 2017). Furthermore, many of these studies have been conducted in non-African populations, raising concerns about their generalizability to Nigerian settings.

Ilorin metropolis, a rapidly urbanizing city in Kwara State, Nigeria, has witnessed an increasing trend in gym-based physical activity among young adults and students. Despite this growing interest in structured exercise, there is a paucity of local data examining the relationship between habitual gym participation and intraocular pressure. Understanding whether regular gym activity influences resting intraocular pressure in this population is essential for developing context-specific recommendations for ocular health.

This study therefore seeks to compare resting intraocular pressure among regular gym-goers and non-regular gym-goers in Ilorin metropolis. By generating locally relevant evidence, the study aims to contribute to the understanding of lifestyle factors that influence intraocular pressure and to support preventive strategies for glaucoma.

1.1 Background of the Study

Intraocular pressure is an important clinical parameter in ophthalmology and optometry, as it reflects the balance between aqueous humour production and drainage within the eye. Under normal physiological conditions, IOP is maintained within a relatively narrow range of approximately 10 to 21 mmHg, although variations may occur due to factors such as age, central corneal thickness, body posture, circadian rhythm, and systemic health status (Bonomi *et al.*, 2000).

The regulation of intraocular pressure is closely linked to ocular perfusion and vascular dynamics. Elevated IOP can compromise ocular blood flow, particularly at the level of the optic nerve head, thereby contributing to the development and progression of glaucomatous optic neuropathy (Flammer *et al.*, 2012). Conversely, adequate ocular perfusion pressure is necessary to maintain the metabolic demands of retinal and optic nerve tissues.

Glaucoma represents a major global health challenge and is a leading cause of irreversible blindness. The global prevalence of glaucoma among adults aged 40 years and above has been estimated at approximately 3.5%, with projections indicating a significant increase in the coming decades (Tham *et al.*, 2025). In Africa, the prevalence is even higher, estimated at about 5.6%, with

primary open-angle glaucoma being the most common subtype (Asiamah *et al.*, 2025). These figures highlight the urgent need for effective preventive and management strategies.

In Nigeria, the burden of glaucoma is substantial, with national surveys reporting that a significant proportion of affected individuals remain undiagnosed until advanced stages of the disease (Kyari *et al.*, 2015). This is largely due to the asymptomatic nature of early glaucoma, limited awareness, and inadequate access to eye care services. As a result, many patients present late with severe visual impairment or blindness.

Given that elevated intraocular pressure is the most important modifiable risk factor for glaucoma, considerable research has focused on identifying factors that influence IOP. These factors include genetic predisposition, systemic diseases such as hypertension and diabetes, and lifestyle factors such as diet, smoking, and physical activity (Weinreb *et al.*, 2014).

Physical exercise has been widely studied for its systemic health benefits, including improved cardiovascular function, enhanced metabolic regulation, and reduced risk of chronic diseases. Exercise induces physiological changes such as increased cardiac output, reduced systemic vascular resistance, and improved endothelial function, all of which may influence ocular blood flow and intraocular pressure (Harris *et al.*, 2008).

Aerobic exercise has been shown to produce transient reductions in intraocular pressure, possibly through mechanisms such as decreased aqueous humour production, increased outflow facility, and reduced episcleral venous pressure (Qureshi *et al.*, 1996). Resistance exercise, on the other hand, may cause temporary increases in IOP due to increased intrathoracic pressure during exertion (Vieira *et al.*, 2006). Despite these findings, the long-term effects of habitual exercise on resting intraocular pressure remain unclear.

Most existing studies have focused on short-term changes in IOP following exercise, rather than baseline measurements that reflect chronic physiological adaptations. Additionally, there is limited evidence from African populations, particularly within Nigerian urban settings such as Ilorin. This gap in knowledge limits the ability of healthcare professionals to provide evidence-based recommendations regarding exercise and ocular health.

1.2 Statement of the Problem

Glaucoma remains a major public health challenge and a leading cause of irreversible blindness in Nigeria. Elevated intraocular pressure is the most important modifiable risk factor associated with the development and progression of glaucoma (Weinreb *et al.*, 2014).

Despite the increasing popularity of gym-based exercise in Ilorin metropolis, there is limited empirical evidence on its long-term effect on resting intraocular pressure. Most available studies have focused on acute post-exercise changes, which may not accurately reflect baseline ocular pressure regulation (Qureshi *et al.*, 1996). As a result, there is insufficient information on whether habitual exercise has a sustained impact on intraocular pressure.

Furthermore, most existing studies have been conducted in non-African populations, limiting their applicability to Nigerian settings. Differences in genetic, environmental, and lifestyle factors may influence the relationship between exercise and intraocular pressure, making it necessary to generate locally relevant data.

The lack of such data poses a challenge for optometrists and other eye care professionals in providing evidence-based lifestyle advice to patients, particularly those at risk of glaucoma. This

study therefore seeks to address this gap by comparing resting intraocular pressure among regular gym-goers and non-regular gym-goers in Ilorin metropolis.

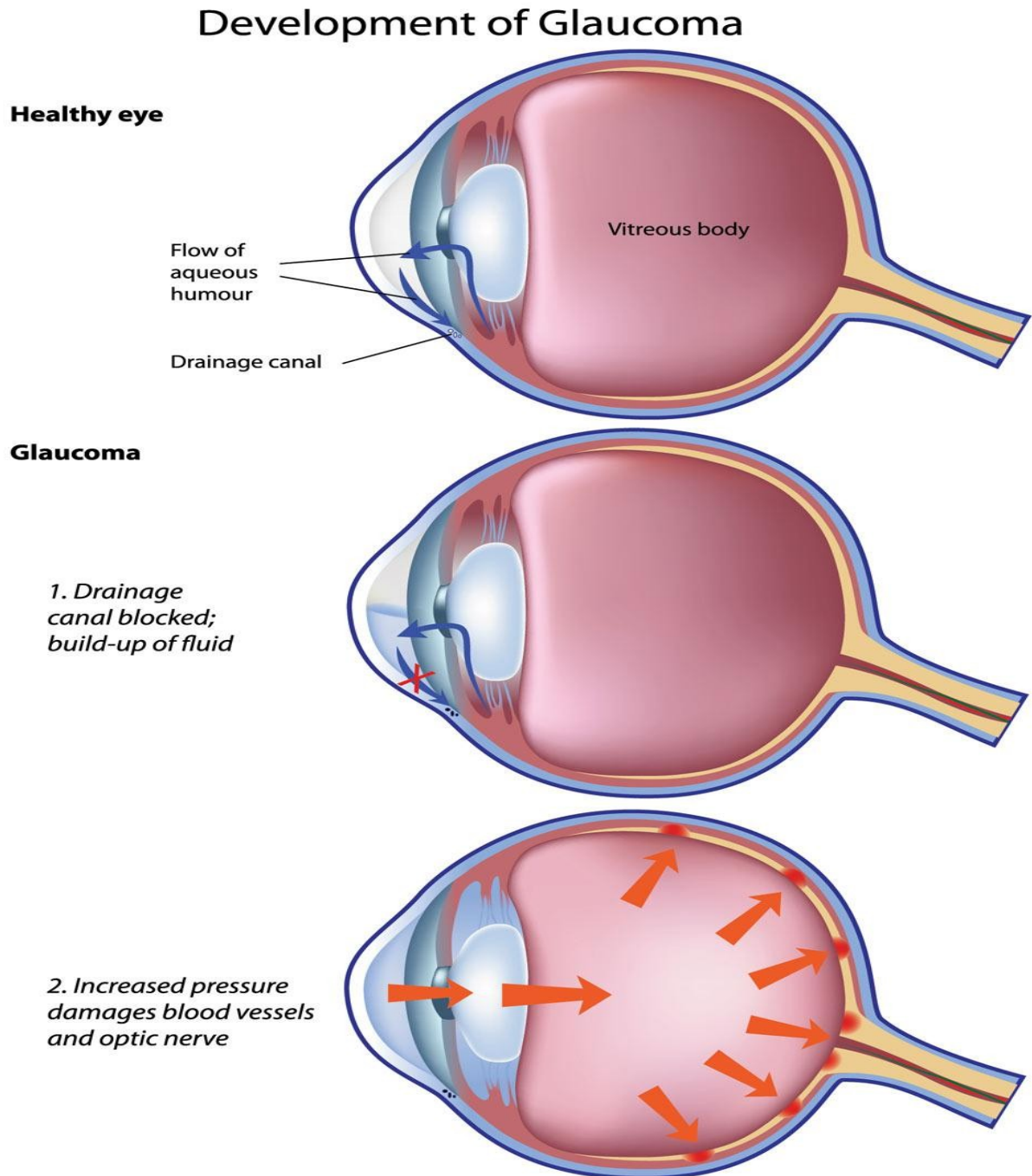


Figure 2: Development of glaucoma showing aqueous humour blockage and optic nerve damage (Adapted from Kanski & Bowling (2016) and Weinreb et al. (2014)).

1.3 Aim of the Study

To compare resting intraocular pressure among regular gym-goers and non-regular gym-goers in Ilorin metropolis.

1.4 Objectives of the Study

1. To determine the mean resting intraocular pressure of regular gym-goers
2. To determine the mean resting intraocular pressure of non-regular gym-goers
3. To compare resting intraocular pressure between both groups
4. To assess the relationship between frequency of gym activity and intraocular pressure

1.5 Research Hypotheses

H_{01} : There is no significant difference in resting intraocular pressure between regular gym-goers and non-regular gym-goers in Ilorin metropolis

H_{02} : There is no significant relationship between frequency of gym activity and intraocular pressure

1.6 Significance of the Study

This study is significant because it contributes to the growing body of knowledge on the relationship between physical exercise and intraocular pressure. The findings will provide insight

into whether habitual gym participation influences resting intraocular pressure, which is a key factor in glaucoma risk assessment.

The results of this study will be useful to optometrists and other eye care professionals in providing evidence-based lifestyle counselling to patients. In particular, the study may inform recommendations regarding physical activity for individuals at risk of glaucoma.

Additionally, the study will generate locally relevant data that can guide future research and public health interventions in Nigeria. By addressing gaps in existing literature, the study will contribute to improved understanding and management of ocular health within the Nigerian context.

1.7 Definition of Terms

Intraocular Pressure (IOP):

Intraocular pressure refers to the hydrostatic pressure exerted by the intraocular fluids within the eye against the ocular walls. It is primarily determined by the balance between aqueous humour production by the ciliary body and its drainage through the trabecular meshwork and uveoscleral pathways. In this study, intraocular pressure is measured in millimetres of mercury (mmHg) and serves as the primary outcome variable.

Resting Intraocular Pressure:

Resting intraocular pressure refers to the baseline intraocular pressure measured under stable physiological conditions without recent physical exertion, emotional stress, or postural changes. It represents the steady-state regulation of ocular pressure and is considered more clinically relevant than transient post-exercise IOP.

Regular Gym-Goer:

A regular gym-goer is defined as an individual who engages in structured physical exercise at least three times per week for a minimum duration of three consecutive months prior to participation in this study. This includes both aerobic and resistance forms of exercise.

Non-Regular Gym-Goer:

A non-regular gym-goer refers to an individual who does not engage in structured or consistent physical exercise, particularly gym-based activities, within the defined frequency and duration specified for regular gym-goers.

Frequency of Gym Activity:

Frequency of gym activity refers to the number of times an individual participates in structured physical exercise within a week. In this study, it is used as a variable to assess the relationship between exercise patterns and intraocular pressure.

Ocular Perfusion Pressure (OPP):

Ocular perfusion pressure is the pressure gradient that facilitates blood flow to the ocular tissues, particularly the optic nerve and retina. It is influenced by systemic blood pressure and intraocular pressure and can be estimated using mean arterial pressure. OPP plays an important role in maintaining ocular health, and reduced perfusion may contribute to optic nerve damage.

Mean Arterial Pressure (MAP):

Mean arterial pressure is the average pressure within the arteries during one cardiac cycle and is calculated using systolic and diastolic blood pressure values. It is used in this study to estimate ocular perfusion pressure and understand its relationship with intraocular pressure.

Aqueous Humour Dynamics:

Aqueous humour dynamics refers to the processes involved in the production, circulation, and drainage of aqueous fluid within the anterior segment of the eye. These processes are critical in maintaining normal intraocular pressure.

Episcleral Venous Pressure:

Episcleral venous pressure refers to the pressure within the venous system responsible for draining blood from the eye. Increased episcleral venous pressure can impede aqueous humour outflow and contribute to elevated intraocular pressure.

Physical Exercise:

Physical exercise refers to planned, structured, and repetitive bodily movement performed to improve or maintain physical fitness, cardiovascular health, and overall well-being. In this study, it includes both aerobic and resistance training performed in a gym setting.

Aerobic Exercise:

Aerobic exercise refers to continuous, rhythmic physical activity that primarily relies on oxygen-dependent energy production, such as running, cycling, or brisk walking. It is generally associated with cardiovascular benefits and potential reductions in intraocular pressure.

Resistance Exercise:

Resistance exercise refers to physical activity involving muscle contraction against an external force, such as weights or resistance bands. It is associated with increased muscular strength and may cause transient increases in intraocular pressure during exertion.

Ilorin Metropolis:

The urban area comprising Ilorin East, Ilorin West, and Ilorin South Local Government Areas of Kwara State, Nigeria.

CHAPTER TWO

LITERATURE REVIEW

2.0 Introduction

Ocular health is an essential component of overall well-being, as vision plays a central role in learning, productivity, and social interaction. Among the various parameters used in assessing ocular health, intraocular pressure (IOP) remains one of the most critical due to its strong association with glaucoma, a leading cause of irreversible blindness worldwide (Weinreb *et al.*, 2014). The regulation of intraocular pressure is therefore of significant clinical importance, particularly in the prevention and management of glaucomatous optic neuropathy.

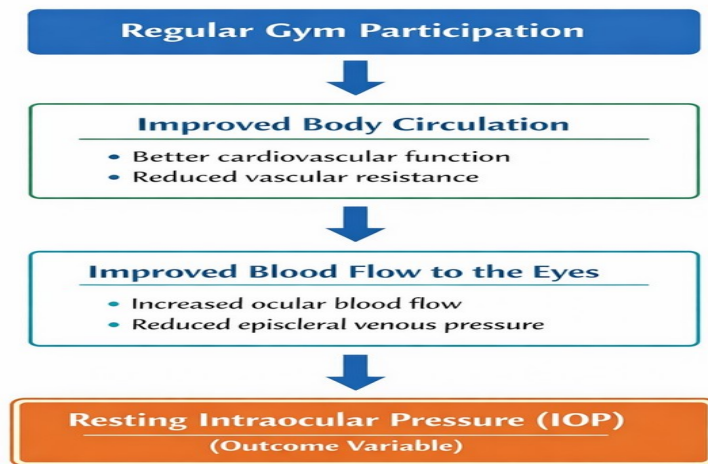
In recent years, increasing attention has been directed toward lifestyle factors that may influence intraocular pressure. Physical exercise, in particular, has emerged as a potential modifiable factor capable of affecting ocular physiology through systemic and local mechanisms. These include cardiovascular adaptation, autonomic nervous system modulation, metabolic regulation, and alterations in ocular fluid dynamics (Yuan *et al.*, 2021; Ma *et al.*, 2022).

Despite the growing body of literature on exercise and intraocular pressure, findings remain inconsistent, especially with regard to long-term effects and resting intraocular pressure. Furthermore, most studies have been conducted in non-African populations, limiting their applicability to Nigerian settings. This chapter therefore reviews relevant literature under conceptual, theoretical, and empirical perspectives, and identifies gaps that justify the present study.

2.1 Conceptual Review

The conceptual framework of this study focuses on the relationship between habitual physical exercise and resting intraocular pressure. The independent variable is regular gym participation, while the dependent variable is resting intraocular pressure. Other variables such as age, gender, blood pressure, and body mass index may influence this relationship

Impact of Regular Exercise on Intraocular Pressure



Fi

Figure 3: Conceptual framework showing the impact of regular exercise on intraocular pressure. (Adapted from Flammer et al. (2012); Harris et al. (2008); and Yuan et al. (2021)).

The framework illustrates the proposed mechanism through which regular gym participation influences intraocular pressure via cardiovascular and ocular perfusion pathways. It highlights the

role of improved circulation and reduced episcleral venous pressure in regulating resting intraocular pressure.

2.1.1 Intraocular Pressure (IOP)

Intraocular pressure is defined as the hydrostatic pressure exerted by intraocular fluids within the eye. It is maintained through a balance between aqueous humour production and its outflow via the trabecular meshwork and uveoscleral pathways. This balance is essential for maintaining ocular shape, optical clarity, and proper functioning of ocular tissues (Kanski & Bowling, 2016).

Normal intraocular pressure typically ranges between 10 and 21 mmHg, although variations may occur due to physiological and environmental factors such as diurnal rhythm, posture, corneal thickness, and systemic conditions (Bonomi *et al.*, 2000). Sustained elevation of IOP can result in optic nerve damage, making it a major risk factor for glaucoma (Weinreb *et al.*, 2014).

2.1.2 Resting Intraocular Pressure

Resting intraocular pressure refers to baseline IOP measured under stable physiological conditions without recent physical exertion or stress. It provides a more accurate representation of long-term ocular pressure regulation compared to transient post-exercise measurements.

Resting IOP is particularly relevant in clinical practice because it reflects the steady-state interaction between aqueous humour dynamics, vascular influences, and autonomic regulation. Changes in resting IOP may therefore indicate underlying physiological adaptations or dysfunction.

2.1.3 Physical Exercise and Intraocular Pressure

Physical exercise involves structured and repetitive bodily movements aimed at improving physical fitness and overall health. Exercise is broadly categorized into aerobic and resistance forms.

Aerobic exercise, such as running and cycling, has been associated with temporary reductions in intraocular pressure. These reductions are thought to occur through mechanisms such as increased ocular blood flow, reduced episcleral venous pressure, and enhanced parasympathetic activity (Qureshi *et al.*, 1996; Vera *et al.*, 2017). In contrast, resistance exercise may lead to short-term increases in IOP due to elevated intrathoracic pressure during exertion, particularly when associated with breath-holding (Vieira *et al.*, 2006).

While these acute effects are well documented, the long-term influence of habitual exercise on resting intraocular pressure remains less clearly understood.

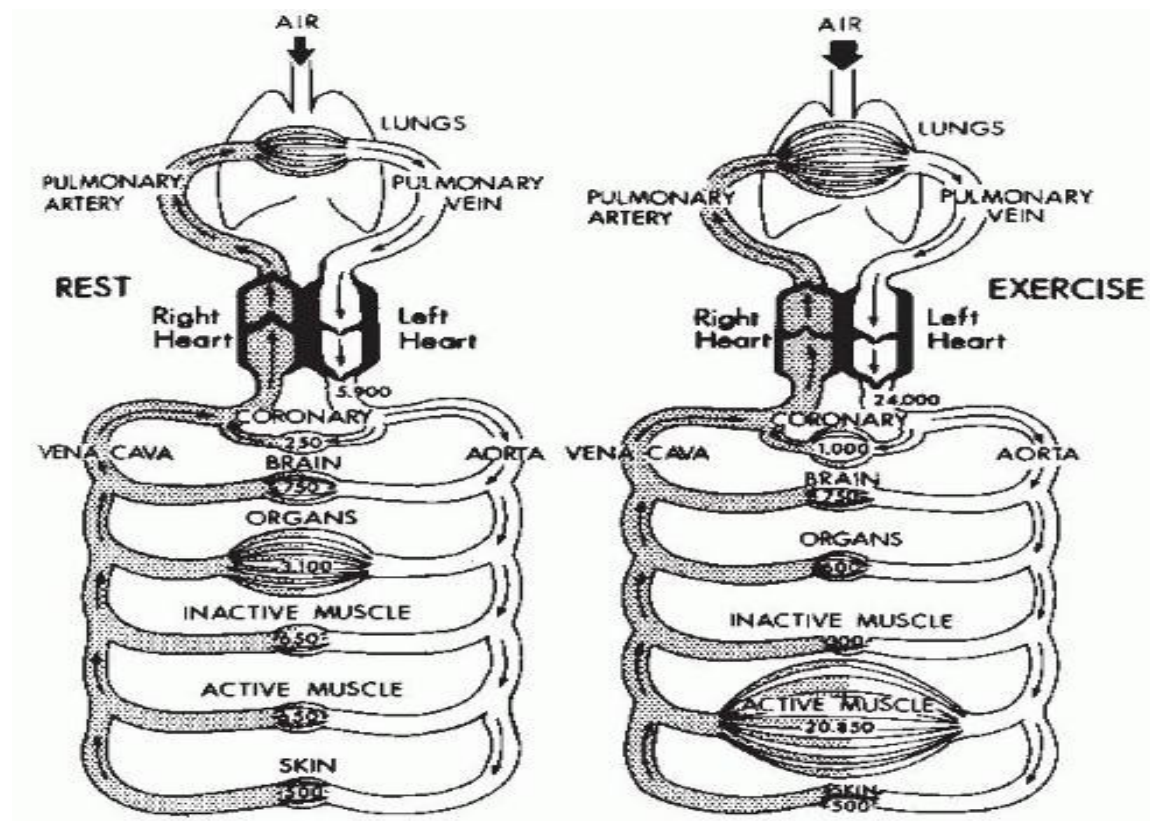


Figure 4: Comparison of systemic blood flow distribution at rest and during exercise
(Adapted from Guyton and Hall (2016)).

This figure illustrates the redistribution of blood flow during exercise, characterized by increased cardiac output and enhanced perfusion to active skeletal muscles, alongside reduced flow to inactive tissues. These cardiovascular adaptations may contribute to improved ocular perfusion and reduced episcleral venous pressure, thereby influencing intraocular pressure regulation.

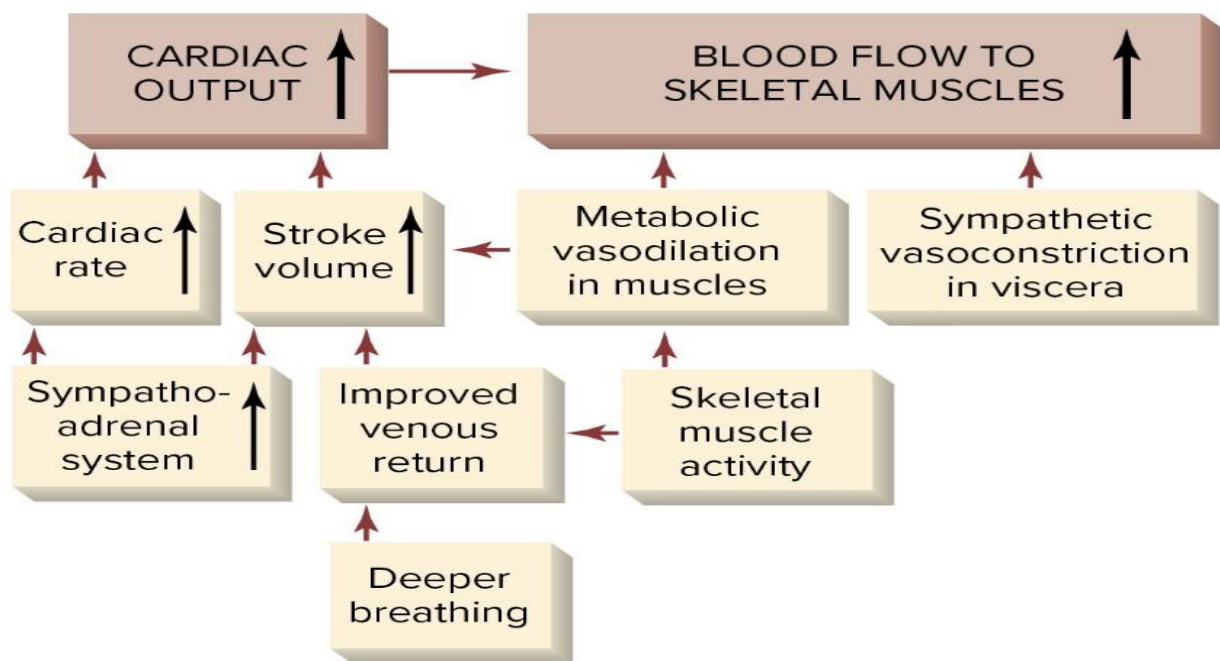


Figure 9.30 A summary of cardiovascular responses to exercise.

Figure 5: Summary of cardiovascular responses to exercise showing increased cardiac output and blood flow redistribution (Adapted from Guyton and Hall (2016)).

2.1.4 Ocular Perfusion Pressure (OPP)

Ocular perfusion pressure is a key physiological parameter that represents the pressure gradient responsible for ocular blood flow. It is influenced by both systemic blood pressure and intraocular pressure.

Ocular Perfusion Pressure Formula

Mean Arterial Pressure (MAP) is calculated as:

$$\text{MAP} = \text{Diastolic Blood Pressure (DBP)} + \frac{1}{3} \times [\text{Systolic Blood Pressure (SBP)} - \text{Diastolic Blood Pressure (DBP)}]$$

Mean Ocular Perfusion Pressure (MOPP) is calculated as:

$$\text{MOPP} = (\frac{2}{3} \times \text{Mean Arterial Pressure (MAP)}) - \text{Intraocular Pressure (IOP)}$$

Substituting for MAP:

$$\text{MOPP} = (\frac{2}{3}) \times [\text{Diastolic Blood Pressure (DBP)} + \frac{1}{3} \times (\text{Systolic Blood Pressure (SBP)} - \text{Diastolic Blood Pressure (DBP)})] - \text{Intraocular Pressure (IOP)}$$

Rearranging to express IOP:

$$\text{IOP} = (\frac{2}{3} \times \text{Mean Arterial Pressure (MAP)}) - \text{Mean Ocular Perfusion Pressure (MOPP)}$$

Mean Arterial Pressure (MAP):

The average pressure within the arteries during one complete cardiac cycle. It reflects the overall blood pressure driving blood through the circulatory system.

Systolic Blood Pressure (SBP):

The maximum pressure in the arteries during ventricular contraction (heart pumping phase).

Diastolic Blood Pressure (DBP):

The minimum pressure in the arteries during ventricular relaxation (heart resting phase).

Mean Ocular Perfusion Pressure (MOPP):

The effective pressure driving blood flow to the ocular tissues, particularly the retina and optic nerve.

Intraocular Pressure (IOP):

The pressure exerted by the intraocular fluids within the eye, maintained by the balance between aqueous humour production and outflow.

This relationship demonstrates that intraocular pressure is closely linked to systemic vascular dynamics. An increase in IOP may reduce ocular perfusion, while improved cardiovascular function may enhance perfusion and support ocular health.

Regular physical exercise may influence IOP indirectly by improving cardiovascular efficiency, reducing systemic vascular resistance, and enhancing blood flow to ocular tissues (Flammer *et al.*, 2012). This highlights the importance of considering OPP when examining the relationship between exercise and intraocular pressure.

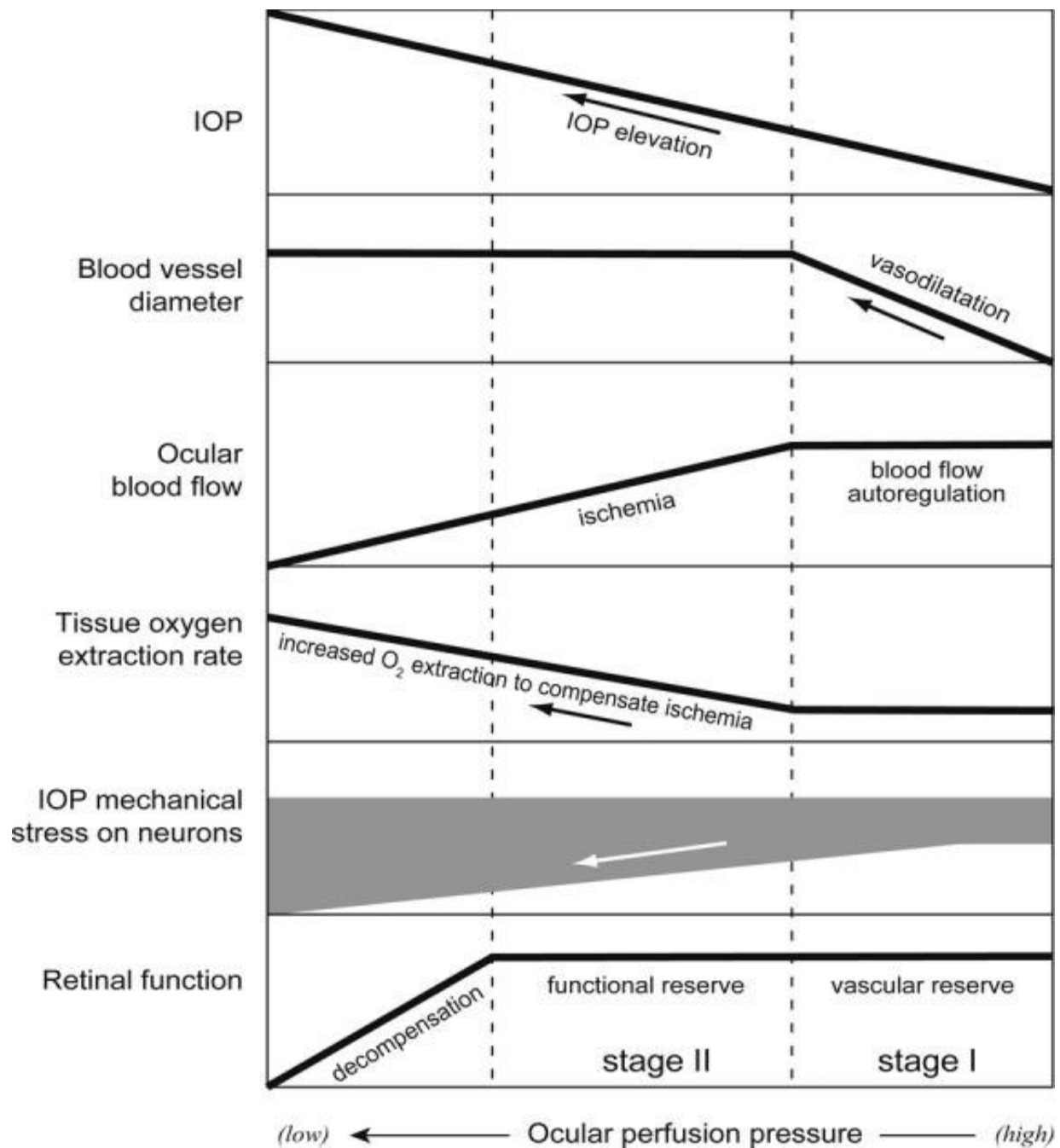


Figure 6: Relationship between intraocular pressure, ocular perfusion pressure, and retinal function (Adapted from Flammer et al. (2012)).

This figure illustrates the relationship between intraocular pressure and ocular perfusion pressure, highlighting the role of vascular autoregulation in maintaining ocular blood flow. As intraocular pressure increases or perfusion pressure decreases, compensatory mechanisms attempt to maintain retinal function. However, prolonged imbalance may lead to ischemia, increased mechanical stress on retinal neurons, and eventual visual dysfunction.

2.1.5 Aqueous Humour Dynamics

Aqueous humour is produced by the ciliary body and flows through the anterior chamber before draining via the trabecular meshwork and uveoscleral pathways. The resistance to outflow within the trabecular meshwork is a major determinant of intraocular pressure.

Factors that influence aqueous humour dynamics include autonomic nervous system activity, vascular pressure, and structural integrity of the outflow pathways. Physical exercise may improve aqueous outflow indirectly by reducing episcleral venous pressure and enhancing vascular function (Yuan *et al.*, 2021).

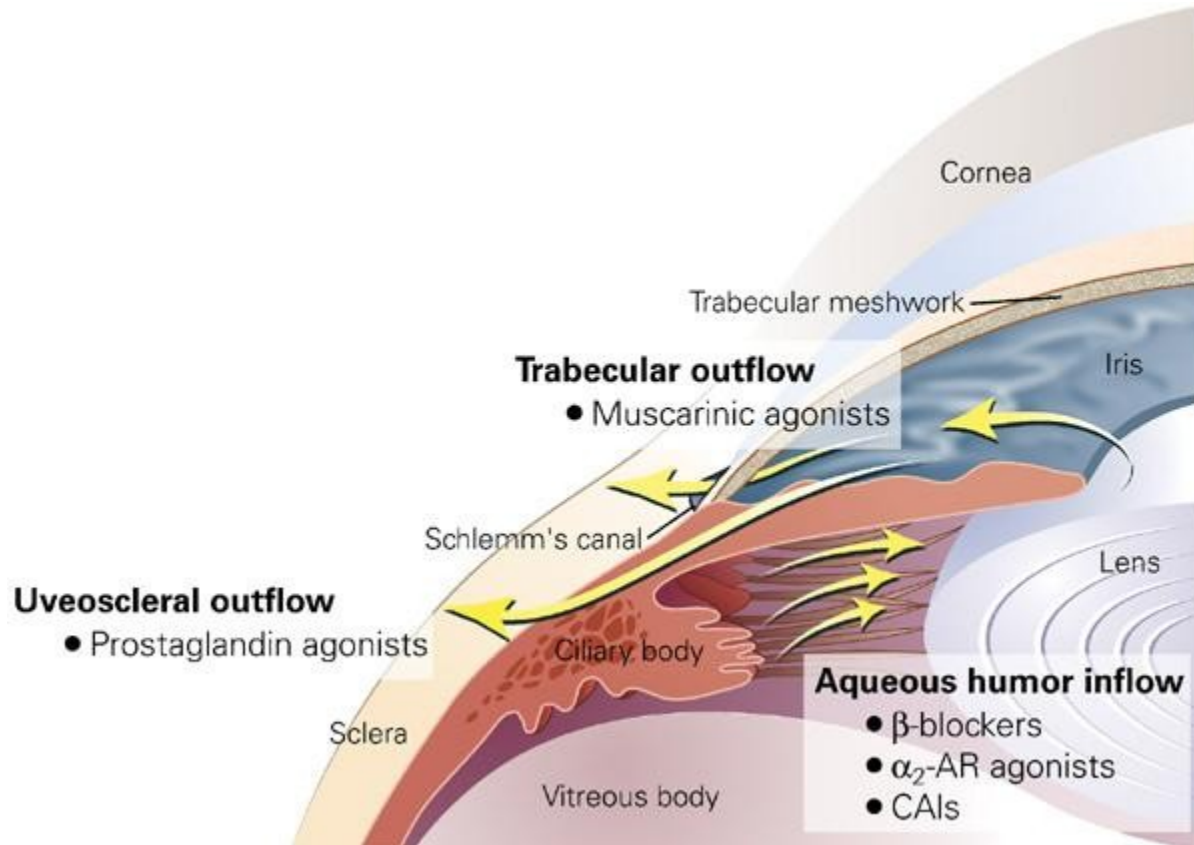


Figure 7: Aqueous humour dynamics showing inflow and outflow pathways of intraocular fluid (Adapted from Kanski and Bowling (2016) and Weinreb et al. (2014)).

2.1.6 Autonomic and Metabolic Regulation

The autonomic nervous system plays a key role in regulating aqueous humour production and ocular blood flow. Sympathetic activation tends to increase aqueous production, while parasympathetic activity promotes outflow.

Regular physical activity enhances parasympathetic tone and reduces sympathetic dominance, creating a physiological environment that may favour optimal intraocular pressure regulation (Chen et al., 2018).

Additionally, exercise improves metabolic health by enhancing insulin sensitivity and reducing systemic inflammation. Chronic inflammation has been associated with dysfunction of the trabecular meshwork, which may contribute to elevated IOP. Thus, the metabolic effects of exercise may indirectly support long-term IOP regulation (Stanescu *et al.*, 2024).

2.2 Theoretical Framework

This study is anchored on three complementary theoretical perspectives: Ocular Perfusion Pressure Theory, Mechanical Theory of Intraocular Pressure Regulation, and Health Lifestyle Theory.

2.2.1 Ocular Perfusion Pressure Theory

This theory posits that intraocular pressure is influenced by systemic vascular dynamics, particularly the balance between blood pressure and vascular resistance. Adequate ocular perfusion is necessary to maintain the health of the optic nerve and retinal tissues.

Exercise improves cardiovascular function, enhances endothelial activity, and reduces vascular resistance, thereby improving ocular perfusion. These adaptations may contribute to better regulation of intraocular pressure (Flammer *et al.*, 2012)

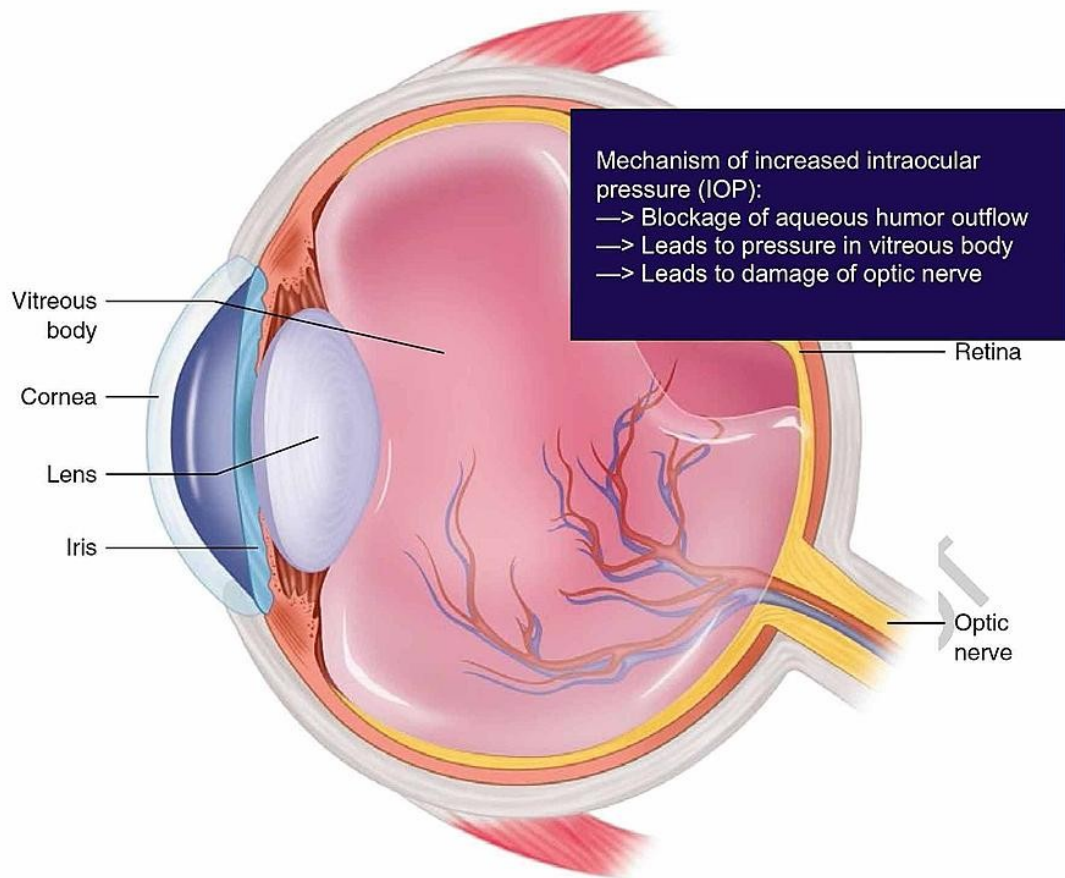


Figure 8: Mechanism of increased intraocular pressure leading to optic nerve damage
(Adapted from Kanski and Bowling (2016) and Weinreb et al. (2014)).

This figure illustrates the mechanism underlying increased intraocular pressure, emphasizing the role of impaired aqueous humour outflow. The resulting accumulation of fluid leads to elevated pressure within the eye, which is transmitted to the optic nerve head, ultimately causing structural damage and progressive visual impairment.

2.2.2 Mechanical Theory of IOP Regulation

The mechanical theory focuses on the balance between aqueous humour production and outflow. Intraocular pressure is determined by resistance within the trabecular meshwork and episcleral venous pressure.

Exercise may influence this balance by reducing venous pressure and improving outflow facility, thereby contributing to lower intraocular pressure under resting conditions.

2.2.3 Health Lifestyle Theory

Health Lifestyle Theory emphasizes the role of habitual behaviours in determining health outcomes. Regular physical activity is considered a positive health behaviour associated with improved cardiovascular, metabolic, and systemic health (Cockerham, 2005).

Individuals who engage in regular exercise may therefore experience physiological adaptations that support better ocular health and optimal intraocular pressure regulation.

2.3 Empirical Review

2.3.1 Acute Effects of Exercise

Moderate-intensity aerobic exercise has consistently been associated with transient reductions in intraocular pressure immediately following activity. These reductions are thought to result from improved ocular blood flow and decreased episcleral venous pressure. Short-term exercise has also been shown to increase ocular perfusion pressure, suggesting improved vascular support for ocular tissues (Yuan *et al.*, 2021; Ma *et al.*, 2022).

However, these effects are typically short-lived, often lasting only minutes to hours after exercise. As such, they do not necessarily reflect long-term changes in baseline intraocular pressure.

2.3.2 Exercise Modality and IOP

Different forms of exercise produce varying effects on intraocular pressure. Aerobic exercise generally leads to reductions in IOP, whereas resistance exercise may cause temporary increases due to elevated intrathoracic pressure during exertion.

These variations highlight the importance of considering exercise type when evaluating the relationship between physical activity and intraocular pressure.

2.3.3 Long-Term Effects of Exercise

Evidence on the long-term effects of exercise on resting intraocular pressure remains limited. Some studies suggest that individuals with higher levels of physical fitness may have lower baseline IOP, possibly due to improved cardiovascular and metabolic function.

However, findings remain inconsistent due to differences in study design, population characteristics, and measurement techniques. This inconsistency underscores the need for further research, particularly in underrepresented populations.

2.3.4 Population and Geographic Limitations

Most studies on exercise and intraocular pressure have been conducted in European and Asian populations. There is limited evidence from African settings, particularly Nigeria, where lifestyle, environmental, and genetic factors may differ significantly.

Urbanization and changing lifestyle patterns in Nigerian cities may influence both physical activity levels and ocular health outcomes, making it important to generate locally relevant data.

2.3.5 Summary of Empirical Evidence

Overall, existing studies suggest that exercise influences intraocular pressure through multiple pathways, including vascular, mechanical, and metabolic mechanisms. While acute reductions in IOP following exercise are well documented, the long-term impact of habitual exercise on resting intraocular pressure remains unclear.

2.4 Research Gaps

From the literature reviewed, several gaps are identified:

- Limited focus on resting intraocular pressure
- Scarcity of studies in African and Nigerian populations
- Lack of differentiation between exercise types
- Inconsistent findings across studies
- Limited integration of theoretical frameworks

2.5 Summary of Literature Review

This chapter has examined the conceptual, theoretical, and empirical literature on the relationship between physical exercise and intraocular pressure. Evidence suggests that exercise influences IOP through multiple physiological pathways, although findings remain inconsistent, particularly with regard to long-term effects.

The identified gaps highlight the need for locally relevant research examining resting intraocular pressure among habitual exercisers. This study therefore aims to address these gaps by comparing IOP among regular gym-goers and non-regular gym-goers in Ilorin metropolis.

CHAPTER THREE

MATERIALS AND METHODS

3.1 Research Design

This study employed a **comparative cross-sectional research design**. This design involves the collection of data from two or more groups at a single point in time for the purpose of comparison. It is particularly suitable for studies that aim to assess differences between groups without manipulating the study variables.

The choice of a comparative cross-sectional design is appropriate for this study because it allows for the direct comparison of intraocular pressure (IOP) between regular gym-goers and non-regular gym-goers under natural conditions. The design also enables the assessment of associations between physical activity and intraocular pressure without the need for long-term follow-up.

Cross-sectional studies are widely used in clinical and epidemiological research due to their efficiency, cost-effectiveness, and ability to provide baseline data on population characteristics (Setia, 2016). However, they do not establish causality, which is acknowledged as a limitation of this study.

3.2 Study Area

The study was conducted in **Ilorin metropolis**, Kwara State, Nigeria. Ilorin is a rapidly growing urban centre comprising Ilorin East, Ilorin West, and Ilorin South Local Government Areas. The city is characterized by a diverse population of students, civil servants, traders, and professionals.

In recent years, Ilorin metropolis has experienced an increase in gym facilities and fitness centres, reflecting a growing interest in physical exercise among residents, particularly young adults. This makes it an appropriate setting for investigating the relationship between habitual gym participation and intraocular pressure.

3.3 Study Population

The study population consisted of **adult residents of Ilorin metropolis aged 18 years and above**.

Participants were categorized into two groups:

- Regular gym-goers
- Non-regular gym-goers

This classification was based on their level of engagement in structured physical exercise.

3.4 Inclusion and Exclusion Criteria

3.4.1 Inclusion Criteria

Participants included in the study met the following criteria:

- Adults aged 18 years and above
- Individuals who provided informed consent
- Regular gym-goers engaging in exercise at least three times per week for a minimum of three months

- Apparently healthy individuals not engaged in regular physical exercise (for control group)

3.4.2 Exclusion Criteria

Participants were excluded if they had:

- Diagnosed glaucoma or ocular hypertension
- History of ocular trauma or surgery
- Use of medications known to affect intraocular pressure (e.g., corticosteroids)
- Systemic conditions that could significantly alter IOP (e.g., severe hypertension)

These criteria were applied to minimize confounding factors that could affect intraocular pressure.

3.5 Sampling Technique

A **multistage sampling technique** was employed for participant selection.

In the first stage, purposive sampling was used to identify gym facilities within Ilorin metropolis. This ensured that participants classified as regular gym-goers were truly engaged in structured exercise programs.

In the second stage, eligible gym-goers were recruited from selected fitness centres using convenience sampling. For the comparison group, non-gym-goers were selected from the general population, including students and residents within the same geographical area.

This combination of purposive and convenience sampling is commonly used in comparative community-based studies where distinct exposure groups are required (Etikan *et al.*, 2016).

3.6 Sample Size Determination

The sample size for this study was determined using the Cochran formula for cross-sectional studies:

Cochran Formula for Sample Size in Cross-Sectional Studies

$$n = \frac{Z^2 pq}{d^2}$$

- ***n*** = required sample size
- ***Z*** = standard normal deviation (1.96 for 95% confidence level)
- ***p*** = estimated proportion of the population with the characteristic of interest
- ***q*** = 1 - ***p***
- ***d*** = margin of error (desired level of precision)

Cochran Formula:

$$n = (Z^2 \times p \times q) / d^2$$

Where:

n = minimum sample size

Z = standard normal deviate (1.96 at 95% confidence level)

p = estimated proportion of the population (0.5)

q = 1 – p = 0.5

d = margin of error (0.05)

Substitution:

$$n = (1.96^2 \times 0.5 \times 0.5) / (0.05^2)$$

Step-by-step calculation:

$$1.96^2 = 3.8416$$

$$0.5 \times 0.5 = 0.25$$

$$d^2 = 0.05^2 = 0.0025$$

Final computation:

$$n = (3.8416 \times 0.25) / 0.0025$$

$$n = 0.9604 / 0.0025$$

$$n = 384.16$$

Approximation:

$$n \approx 384$$

Adjustment for non-response (10%):

10% of 384 = 38.4

Adjusted sample size:

$n = 384 + 38.4$

$n = 422.4$

$n \approx 422$

Final Sample Used:

However, due to time, financial, and logistical constraints, a total of **120 participants** were used for this study, comprising:

- 60 regular gym-goers
- 60 non-regular gym-goers

3.7 Instruments for Data Collection

3.7.1 Equipment

The following instruments were used:

- **Non-contact Tonometer (NCT):** Used for measuring intraocular pressure
- **Pen torch:** For basic external ocular examination
- **Stopwatch/clock:** For timing procedures
- **Data recording sheets**
- **Weighing scale**

- **Automated Blood Pressure Monitor**

The tonometer is considered a standard and reliable instrument for measuring intraocular pressure in clinical settings (Weinreb *et al.*, 2014).

3.7.2 Questionnaire

A structured questionnaire was used to collect data on:

- Socio-demographic characteristics
- Physical activity patterns
- Frequency and duration of gym participation

Questionnaires are widely used in epidemiological studies due to their efficiency and ability to capture large amounts of data (Bowling, 2014).

3.8 Procedure for Data Collection

3.8.1 Ethical Consideration

Ethical approval was obtained from the appropriate institutional review board before commencement of the study. Participants were informed about the purpose of the study, and written informed consent was obtained prior to participation.

Confidentiality of participant information was strictly maintained, and participation was voluntary.

3.8.2 Pre-Examination Procedure

Participants were advised to:

- Avoid strenuous exercise prior to measurement
- Maintain a relaxed state before IOP assessment

This was done to ensure that measured values represented **resting intraocular pressure**.

3.8.3 Measurement of Intraocular Pressure

Intraocular pressure was measured using a non-contact tonometer under standardized conditions.

Participants were seated comfortably, and care was taken to avoid external pressure on the eye.

Measurements were taken at consistent times where possible to minimize diurnal variation.

Multiple readings were taken, and the average value was recorded to improve accuracy.

3.8.4 Data Recording

All measurements and responses were recorded immediately on data sheets and later transferred into a digital database for analysis.

3.9 Validity and Reliability of Instruments

The tonometer used for measuring intraocular pressure is a validated clinical instrument widely accepted in ophthalmic practice. Calibration was performed prior to use to ensure accuracy.

The questionnaire was pre-tested to ensure clarity, relevance, and consistency. This helped to minimize ambiguity and improve reliability of responses.

3.10 Method of Data Analysis

Data collected were entered, cleaned, and analysed using:

- SPSS

Descriptive statistics were used to summarize data:

- Mean
- Standard deviation
- Frequency and percentages

Inferential statistics included:

- **Independent samples t-test** → to compare IOP between groups
- **Pearson correlation** → to assess relationship between exercise frequency and IOP

Statistical significance was set at: $p < 0.05$

3.11 Limitations of the Study

This study is subject to certain limitations:

- The cross-sectional design does not establish causality
- Self-reported exercise data may introduce recall bias

- Limited sample size compared to calculated Cochran value
- Study confined to Ilorin metropolis, limiting generalizability

CHAPTER FOUR

DATA ANALYSIS AND RESULTS

4.0 Overview

A total of 120 participants were enrolled in the study, comprising 60 regular gym-goers and 60 non-regular gym-goers. The results are presented sequentially in line with the study's objectives: demographic characterization of the sample, assessment of the normality of IOP distributions, descriptive analysis of IOP within each group, inferential comparison of IOP between groups, effect size estimation, and correlation analysis between exercise frequency and resting IOP. Statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS), version 26.0. The threshold for statistical significance was set at $p < .05$.

4.1 Demographic and Baseline Characteristics of Participants

The demographic and baseline clinical characteristics of participants in both groups are summarized in **Table 1**. The two groups were comparable with respect to mean age, systolic blood pressure, diastolic blood pressure, and body mass index (BMI), lending confidence to the conclusion that observed differences in IOP are attributable to gym participation rather than pre-existing physiological differences between groups.

Table 1: Demographic and Baseline Characteristics of Participants by Group

Variable	Gym Goer (n = 60)	Non-Gym Goer (n = 60)	Total (N = 120)
Age — Mean (SD)	23.57 (3.68)	22.37 (3.08)	22.97 (3.43)
Systolic BP — Mean (SD)	120.38 (7.99)	120.12 (10.21)	120.25 (9.13)
Diastolic BP — Mean (SD)	78.25 (7.16)	78.65 (9.60)	78.45 (8.44)
BMI — Mean (SD)	24.44 (3.18)	23.48 (3.66)	23.96 (3.45)
Gender: Male n (%)	38 (63.3%)	22 (36.7%)	60 (50.0%)
Gender: Female n (%)	22 (36.7%)	38 (63.3%)	60 (50.0%)

Note. BP = blood pressure; BMI = body mass index; SD = standard deviation.

The gym-goer group had a slightly higher mean age (23.57 ± 3.68 years) compared to the non-gym-goer group (22.37 ± 3.08 years). Mean systolic and diastolic blood pressure values were virtually identical across groups, as were BMI values, suggesting that cardiovascular and anthropometric profiles were balanced at baseline. With respect to gender distribution, the gym-goer group was predominantly male (63.3%), while the non-gym-goer group was predominantly female (63.3%), reflecting typical gym attendance patterns in the Ilorin metropolis setting where male gym participation tends to be higher.

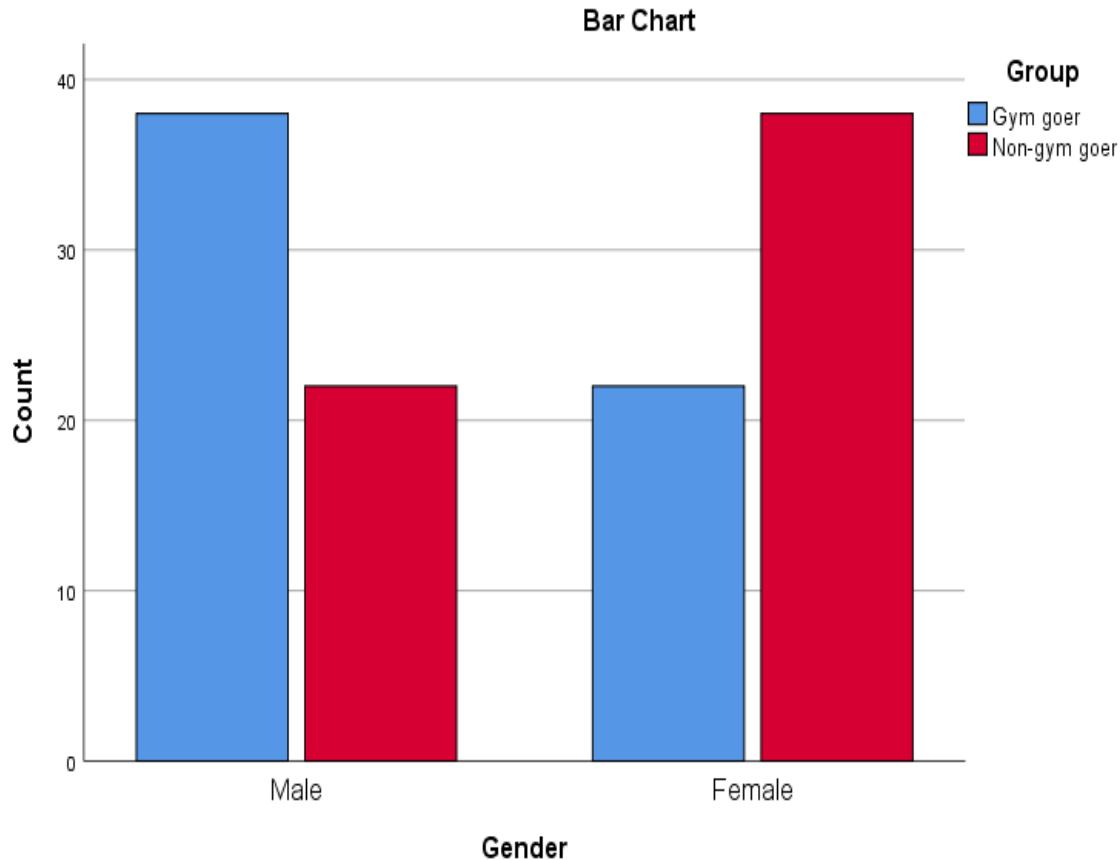


Figure 9. Gender distribution of participants by group.

Figure 9 illustrates the gender composition of the gym-goer and non-gym-goer groups. Of the 60 gym-goers, 38 (63.3%) were male and 22 (36.7%) were female. The non-gym-goer group showed the inverse pattern, with 22 males (36.7%) and 38 females (63.3%). The total sample was gender-balanced overall (60 males, 60 females; 50% each), but the within-group gender distribution was notably asymmetric between the two groups. This pattern likely reflects the generally higher rates of gym attendance among males in the Ilorin metropolitan context, which is consistent with trends reported in physical activity participation studies across sub-Saharan Africa.

This gender imbalance between groups is acknowledged as a potential confounding variable, as some studies have reported modest gender-related differences in intraocular pressure (with males

tending toward slightly higher IOP in certain age brackets). However, since the overall sample was gender-balanced and both groups fell within the normal IOP range, this imbalance is noted as a limitation rather than a disqualifying factor and is addressed in the limitations section of *Chapter Five*.

4.2 Assessment of Normality of IOP Distributions

Prior to conducting inferential analyses, the normality of resting IOP distributions was formally examined for both groups using the Shapiro-Wilk test, which is the preferred normality test for samples of 300 or fewer (Razali & Wah, 2011; Pallant, 2020). The Shapiro-Wilk test was supplemented by examination of skewness and kurtosis statistics, as well as visual inspection of histograms and normal Q-Q plots. The results of these tests are presented in Table 2.

Table 2: Tests of Normality for Resting IOP by Group (Shapiro-Wilk)

Variable	Group	W	df	p	Interpretation
IOP_Avg	Gym goer	.992	60	.955	Normal ✓
	Non-gym goer	.977	60	.315	Normal ✓
IOP_R	Gym goer	.985	60	.652	Normal ✓
	Non-gym goer	.976	60	.279	Normal ✓
IOP_L	Gym goer	.990	60	.918	Normal ✓
	Non-gym goer	.971	60	.159	Normal ✓

Note. All Shapiro-Wilk p-values > .05, confirming normality assumption. W = Shapiro-Wilk statistic.

Shapiro-Wilk results confirmed that IOP_Avg was normally distributed in both the gym-goer group ($W = .992$, $p = .955$) and the non-gym-goer group ($W = .977$, $p = .315$). Similarly, right eye IOP (IOP_R) and left eye IOP (IOP_L) were normally distributed in both groups (all $p > .05$). These findings were corroborated by skewness and kurtosis values that remained well within the acceptable range of ± 1 for both groups across all three IOP measures, indicating minimal departure from symmetry and no problematic tail behavior. Visual inspection of histograms and normal Q-Q plots likewise confirmed approximate normality.

Given that the assumption of normality was satisfied for both groups, the use of the parametric independent-samples t-test for the primary group comparison (Objective 3) was fully justified. No transformation of IOP data was required prior to analysis.

4.3 Descriptive Statistics of Resting Intraocular Pressure (Objectives 1 & 2)

Objectives 1 and 2 sought to determine the mean resting IOP of gym-goers and non-gym-goers, respectively, in Ilorin metropolis. Descriptive statistics for average IOP (IOP_Avg), right eye IOP (IOP_R), and left eye IOP (IOP_L) are presented in Table 3.

Table 3: Descriptive Statistics for Resting IOP by Group

Measure	Group	Mean	SD	Median	Min	Max	Skew
IOP_Avg (mmHg)	Gym goer	14.82	2.30	15.01	10.00	21.15	.10
	Non-gym goer	16.73	3.22	16.32	10.52	24.20	.35
IOP_R (mmHg)	Gym goer	14.88	2.36	14.70	9.90	20.30	.03
	Non-gym goer	16.81	3.46	16.60	10.70	25.30	.39
IOP_L (mmHg)	Gym goer	14.76	2.38	14.75	9.10	22.00	.16
	Non-gym goer	16.66	3.24	16.10	10.10	25.00	.47

Note. IOP = intraocular pressure; SD = standard deviation; Skew = skewness. All values in mmHg.

Objective 1 — Gym-goers: The mean resting IOP_Avg for regular gym-goers was 14.82 ± 2.30 mmHg (95% CI: 14.23–15.42). The median value (15.01 mmHg) was close to the mean, reinforcing the symmetry of the distribution. IOP values ranged from 10.00 to 21.15 mmHg, with

an interquartile range of 3.16 mmHg. Right eye and left eye IOP were virtually identical in the gym-goer group (IOP_R: 14.88 mmHg; IOP_L: 14.76 mmHg), indicating consistent bilateral IOP patterns in this group.

Objective 2 — Non-gym-goers: The mean resting IOP_Avg for non-regular gym-goers was 16.73 $\pm$ 3.22 mmHg (95% CI: 15.90–17.56). The wider standard deviation and interquartile range (4.31 mmHg) indicate greater variability in resting IOP within the non-gym-goer group compared to gym-goers. The range extended from 10.52 to 24.20 mmHg. Right eye (16.81 mmHg) and left eye (16.66 mmHg) values were again consistent with the average IOP measure.

Across both groups, mean IOP values fall within the widely accepted population normal range of 10–21 mmHg (Quigley & Broman, 2006; Bourne *et al.*, 2016), indicating that neither group exhibited ocular hypertension at baseline. However, a clear numerical difference of approximately 1.91 mmHg is apparent between groups, favoring lower resting IOP in the gym-goer group — a difference subjected to formal inferential testing in the following section.

The extreme value analysis identified five highest and five lowest IOP_Avg values per group. For gym-goers, the highest recorded IOP_Avg was 21.15 mmHg (Participant 48), which, while at the upper end of the normal range, did not constitute a clinical outlier. For non-gym-goers, the highest IOP_Avg was 24.20 mmHg (Participant 27), which technically exceeds the commonly cited 21 mmHg upper limit of normal. These observations are noted but do not alter the conclusions of the analysis, as they represent single-measurement values for individual participants without a confirmed diagnosis of ocular hypertension.

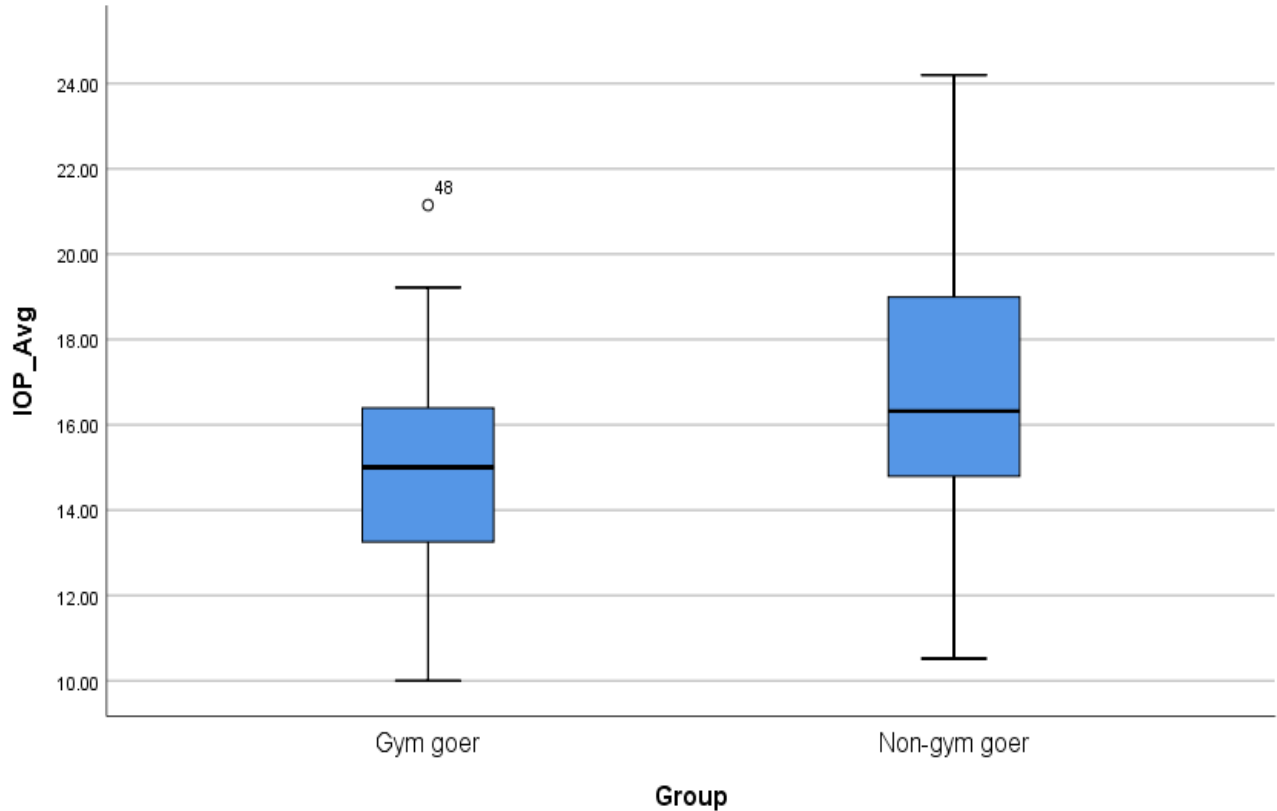


Figure 10 Boxplot of mean resting intraocular pressure (IOP_Avg) by group.

Figure 10 displays the distribution of resting IOP_Avg for gym-goers and non-gym-goers. The gym-goer group demonstrated a lower median IOP (15.01 mmHg) with a narrower interquartile range (IQR = 3.16 mmHg), reflecting greater homogeneity within the group. In contrast, the non-gym-goer group showed a higher median IOP (16.32 mmHg) and a wider IQR (4.31 mmHg), indicating greater dispersion. One mild outlier was identified in the gym-goer group (Participant 48; IOP_Avg = 21.15 mmHg), which falls at the upper boundary of the accepted normal IOP range but does not represent a pathological value. The visual separation between the two boxes supports the statistically significant group difference subsequently confirmed by the independent-samples t-test.

4.4 Comparison of Mean Resting IOP Between Groups — Independent-Samples T-Test (Objective 3 / H_{01})

Objective 3 sought to compare mean resting IOP between regular gym-goers and non-regular gym-goers. The null hypothesis H_{01} stated that there would be no significant difference in resting IOP between the two groups. An independent-samples t-test was employed, with Levene's test for equality of variances conducted as a prerequisite to determine which row of the t-test output was appropriate to report.

4.4.1 Levene's Test for Equality of Variances

Levene's test yielded a statistically significant result ($F = 4.693$, $p = .032$), indicating that the variances of IOP_Avg differed significantly between the two groups. This finding is consistent with the descriptive data, which showed a notably higher standard deviation in the non-gym-goer group ($SD = 3.22$) compared to the gym-goer group ($SD = 2.30$), yielding a variance ratio of approximately 1.95. As the equal variances assumption was violated, the Welch-adjusted t-test row ('Equal variances not assumed') was used for all subsequent reporting, in line with standard statistical practice (Pallant, 2020; Field, 2018).

4.4.2 T-Test Results

Table 4: Independent-Samples T-Test Results for IOP_Avg

	F	Levene p	t	df	p (2- tail)	Mean Diff.	95% CI
<i>Equal variances assumed</i>	4.693	.032*	-3.741	118	.000	-1.910	-2.922 to -0.899
Equal variances NOT assumed (reported)	—	—	-3.741	106.885	.000***	-1.910	-2.923 to -0.898

Note. Highlighted row = row reported. * Levene's $p < .05 \rightarrow$ equal variances not assumed. *** $p < .001$.

The independent-samples t-test using the Welch-adjusted degrees of freedom revealed a statistically significant difference in mean resting IOP between regular gym-goers and non-regular gym-goers: $t(106.885) = -3.741, p < .001$. The mean IOP_Avg of gym-goers (14.82 ± 2.30 mmHg) was significantly lower than that of non-gym-goers (16.73 ± 3.22 mmHg), with a mean difference of -1.91 mmHg (95% CI: -2.92 to -0.90 mmHg). On this basis, the null hypothesis H_{01} is rejected.

Decision on H_{01} : REJECTED — There is a statistically significant difference in resting intraocular pressure between regular gym-goers and non-regular gym-goers in Ilorin metropolis ($p < .001$). Gym-goers had significantly lower resting IOP.

4.5 Effect Size Estimation — Cohen's d and Hedges' g

Statistical significance alone does not convey the practical or clinical importance of a difference; a finding can be statistically significant with minimal real-world relevance if the sample is sufficiently large. Effect size statistics quantify the magnitude of the observed difference independent of sample size, and their reporting is now standard practice in health research (Cohen, 1988; Lakens, 2013). Two complementary effect size indices were computed: Cohen's d and Hedges' g.

4.5.1 Cohen's d

Cohen's d was calculated using the pooled standard deviation of the two groups. The pooled SD was derived as follows:

$$\text{Pooled SD} = \sqrt{[(n_1 - 1) \times SD_1^2 + (n_2 - 1) \times SD_2^2] / (n_1 + n_2 - 2)}$$

$$\text{Pooled SD} = \sqrt{[(59 \times 5.299) + (59 \times 10.344)] / 118}$$

$$\text{Pooled SD} = \sqrt{[312.64 + 610.30] / 118}$$

$$\text{Pooled SD} = \sqrt{[7.819]} = 2.797 \text{ mmHg}$$

$$\text{Cohen's } d = |Mean_1 - Mean_2| / \text{Pooled SD}$$

$$\text{Cohen's } d = |14.824 - 16.734| / 2.797$$

$$\text{Cohen's } d = 1.910 / 2.797 = 0.68$$

4.5.2 Hedges' g (Small-Sample Correction)

Hedges' g applies a correction factor for small-to-moderate sample sizes that renders d slightly more conservative and less biased. The correction factor J is:

$$J = 1 - [3 / (4 \times (n_1 + n_2 - 2) - 1)]$$

$$J = 1 - [3 / (4 \times 118 - 1)]$$

$$J = 1 - [3 / 471] = 1 - 0.00637 = 0.9936$$

$$\text{Hedges' } g = d \times J = 0.683 \times 0.994 = 0.679$$

The near-identical values of Cohen's d (0.68) and Hedges' g (0.68) confirm the stability of the effect size estimate. The effect size indices for all three IOP measures are summarized in **Table 5**.

Table 5: Summary of Effect Size Estimates for Group Differences in IOP

IOP Measure	Mean Gym (SD)	Mean Non-Gym (SD)	Pooled SD	Cohen's d	Interpretation
IOP_Avg	14.82 (2.30)	16.73 (3.22)	2.80	0.68	Medium
IOP_R	14.88 (2.36)	16.81 (3.46)	2.96	0.65	Medium
IOP_L	14.76 (2.38)	16.66 (3.24)	2.84	0.67	Medium

Note. Effect size interpretation based on Cohen (1988): small = 0.20, medium = 0.50, large = 0.80.

Cohen's d of 0.68 for IOP_Avg, corroborated by nearly identical values for IOP_R ($d = 0.65$) and IOP_L ($d = 0.67$), falls within the medium effect size range using Cohen's (1988) widely cited benchmarks (small ≥ 0.20 ; medium ≥ 0.50 ; large ≥ 0.80). This indicates that the observed difference in resting IOP between gym-goers and non-gym-goers is not only statistically significant but is of meaningful practical and clinical magnitude. A medium effect size in a clinical IOP context is noteworthy, as the absolute mean difference of approximately 1.91 mmHg is clinically relevant — several landmark glaucoma studies have demonstrated that IOP reductions of 1–2 mmHg can significantly alter the risk trajectory of glaucomatous disease progression (Heijl *et al.*, 2002; Kass *et al.*, 2002).

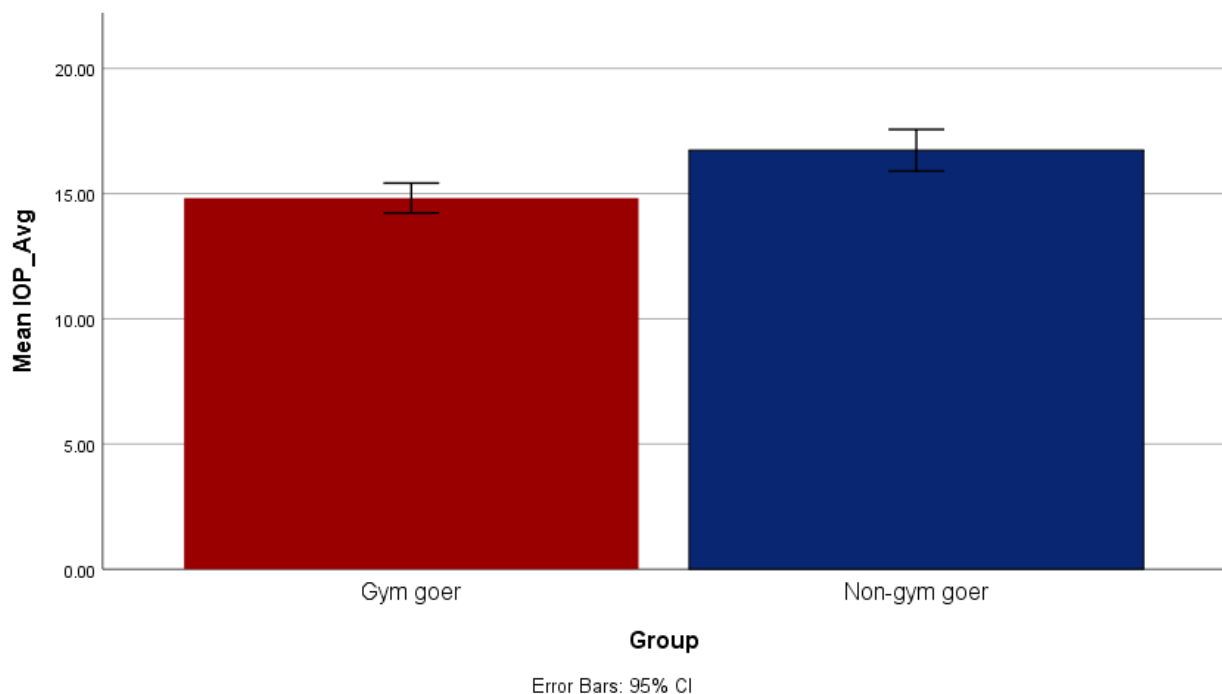


Figure 11: Mean resting IOP_Avg by group with 95% confidence interval error bars.

Figure 11 illustrates the mean resting IOP_Avg for gym-goers ($M = 14.82$ mmHg) and non-gym-goers ($M = 16.73$ mmHg), with error bars representing 95% confidence intervals. The non-

overlapping confidence intervals between the two groups provide a visual confirmation of the statistically significant difference detected by the independent-samples t-test ($t(106.885) = -3.741$, $p < .001$). The mean difference of 1.91 mmHg, while appearing modest on the scale, is clinically relevant in the context of intraocular pressure management and glaucoma risk.

4.6 Relationship Between Exercise Frequency and Resting IOP (Objective 4 / H_{02})

Objective 4 examined whether a significant association exists between frequency of gym activity and resting IOP among gym-goers. The null hypothesis H_{02} stated that there would be no significant association between exercise frequency and resting IOP. Both Pearson and Spearman correlation analyses were conducted within the gym-goer subgroup ($n = 60$) to assess this relationship, with the analysis filtered to gym-goers exclusively.

Table 6: Pearson and Spearman Correlation Between Exercise Frequency and IOP_Avg (Gym-Goers, $n = 60$)

Test	Coefficient	p (2-tailed)	Interpretation
Pearson r	.139	.288	Not significant
Spearman's ρ	.135	.304	Not significant

Both Pearson correlation ($r = .139$, $p = .288$) and Spearman's rho ($\rho = .135$, $p = .304$) revealed weak, positive, and statistically non-significant associations between exercise frequency and

resting IOP among gym-goers. Neither coefficient approached statistical significance at the $p < .05$ threshold, and the magnitude of both coefficients ($< .20$) indicates a negligible to weak relationship at best. The null hypothesis H_{02} is therefore retained.

Decision on H_{02} : RETAINED — There is no statistically significant association between frequency of gym activity and resting IOP among gym-goers in Ilorin metropolis (Pearson $r = .139$, $p = .288$; Spearman $\rho = .135$, $p = .304$).

This finding suggests that while gym participation as a whole is associated with significantly lower resting IOP (as demonstrated by the t-test), the frequency of exercise sessions within the gym-going group does not modulate IOP further in a dose-dependent manner — at least not within the range of frequencies captured in this sample (1–2x to daily per week). This is plausible if the IOP-lowering effect of habitual exercise reaches a plateau or ceiling at the level of regular moderate engagement, such that additional sessions beyond a baseline frequency do not confer proportionally greater IOP reduction.

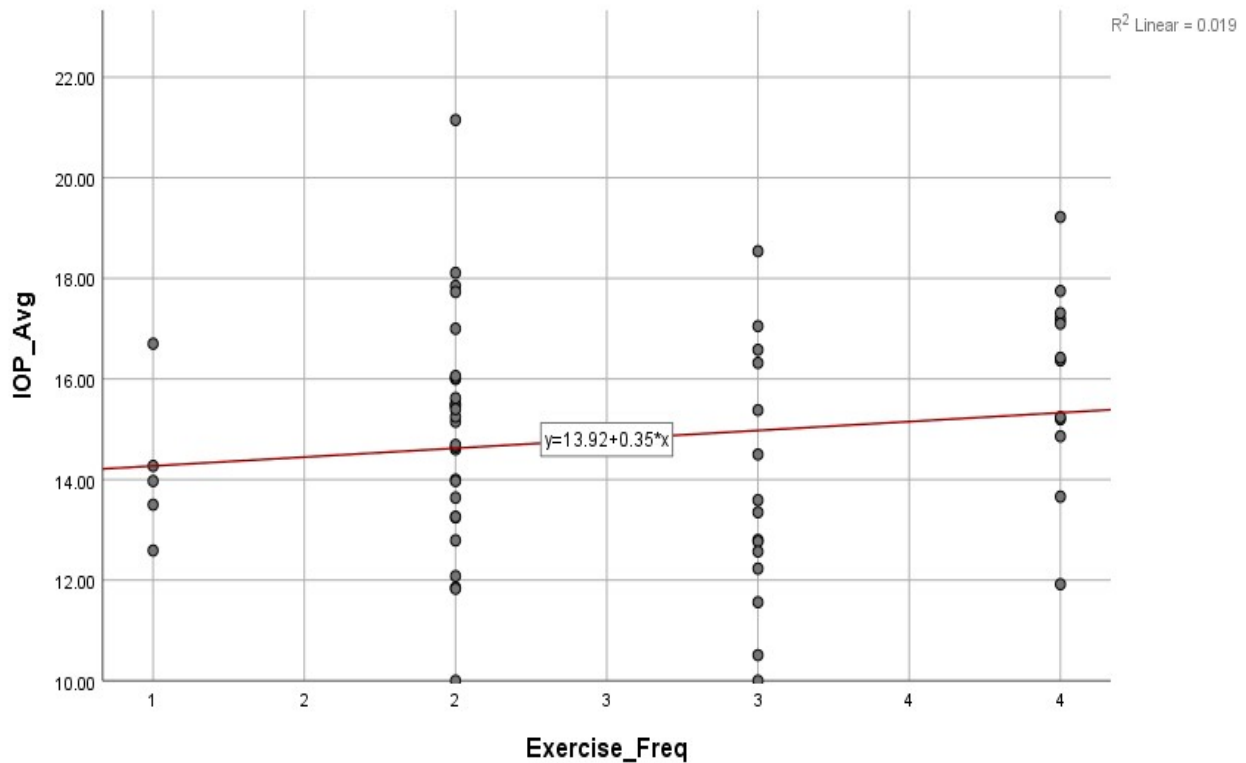


Figure 12. Scatterplot of exercise frequency against mean resting IOP_Avg among gym-goers (n = 60) with linear regression fit line.

Figure 12 presents the relationship between exercise frequency (coded 1 = 1–2×/week, 2 = 3–4×/week, 3 = 5–6×/week, 4 = Daily) and resting IOP_Avg within the gym-goer subgroup (n = 60). The overlaid linear regression line ($y = 13.92 + 0.35x$) shows a marginally positive slope, suggesting a very slight tendency for IOP to increase with higher exercise frequency categories — however, this trend is neither statistically significant nor clinically meaningful. The coefficient of determination ($R^2 = 0.019$) indicates that exercise frequency accounts for only 1.9% of the variance in resting IOP among gym-goers, consistent with the non-significant Pearson correlation ($r = .139$, $p = .288$) and Spearman's rho ($\rho = .135$, $p = .304$) reported in Table 6. The wide scatter of data

points around the fit line further confirms the absence of a meaningful dose-response relationship between exercise frequency and resting IOP in this sample.

CHAPTER FIVE

DISCUSSIONS, CONCLUSION AND RECOMMENDATIONS

5.1 Discussion

The study was motivated by the significant public health burden of glaucoma — the leading cause of irreversible blindness globally — and by the established, but insufficiently studied, relationship between habitual physical activity and IOP in a sub-Saharan African urban population.

A total of 120 participants were recruited using a multistage sampling approach: 60 regular gym-goers identified through purposive sampling from gym facilities in Ilorin metropolis, and 60 non-regular gym-goers recruited from the general population using convenience sampling. Inclusion criteria required gym-goers to be adults aged 18 years and above who engaged in structured physical exercise at least three times per week for a minimum of three months. Individuals with a known history of glaucoma or ocular hypertension, those on medications known to affect IOP, and those with a history of ocular trauma or surgery were excluded from both groups.

Resting IOP was measured using a non-contact tonometer (NCT) under standardized conditions, with participants seated comfortably and standardized protocols observed to minimize measurement variability. Sociodemographic and physical activity data were collected via a structured questionnaire. All data were entered and analyzed in SPSS version 26.0. Statistical methods included descriptive statistics (means, standard deviations, confidence intervals, and distribution parameters), the Shapiro-Wilk test of normality, the independent-samples t-test with

Levene's test for equality of variances, effect size estimation (Cohen's d and Hedges' g), and Pearson and Spearman bivariate correlation analysis.

The principal findings of the study may be summarized as follows:

- The mean resting IOP of regular gym-goers in Ilorin metropolis was **14.82 ± 2.30 mmHg** (95% CI: 14.23–15.42 mmHg), which falls within the accepted normal IOP range.
- The mean resting IOP of non-regular gym-goers was **16.73 ± 3.22 mmHg** (95% CI: 15.90–17.56 mmHg), also within the normal range but numerically and significantly higher than that of gym-goers.
- The independent-samples t -test revealed a statistically significant difference in resting IOP between the two groups ($t(106.885) = -3.741$, $p < .001$, mean difference = -1.91 mmHg, 95% CI: -2.92 to -0.90). The null hypothesis H_{01} was rejected.
- The effect size was medium in magnitude (Cohen's $d = 0.68$; Hedges' $g = 0.68$), indicating that the observed IOP difference between groups is not only statistically significant but clinically meaningful.
- Correlation analysis within the gym-goer subgroup showed no statistically significant association between exercise frequency and resting IOP (Pearson $r = .139$, $p = .288$; Spearman $\rho = .135$, $p = .304$). The null hypothesis H_{02} was retained.

5.1.1 Lower Resting IOP in Gym-Goers: Mechanisms and Literature Context

The finding of significantly lower mean resting IOP in regular gym-goers (14.82 mmHg) compared to non-gym-goers (16.73 mmHg) is consistent with a substantial body of literature

documenting an inverse association between habitual physical activity and intraocular pressure. Several physiological mechanisms have been proposed to explain this relationship.

Firstly, regular aerobic and resistance exercise is associated with improvements in aqueous humor dynamics — the primary determinant of IOP homeostasis. Specifically, habitual exercise has been hypothesized to reduce aqueous humor production through reduced sympathetic tone or to enhance trabecular outflow facility, thereby lowering the steady-state IOP (Risner *et al.*, 2009; Roddy *et al.*, 2021). Trabecular meshwork cells are known to be mechanosensitive, and repeated mechanical stimulation through regular exercise may upregulate outflow pathway efficiency over time.

Secondly, the cardiovascular benefits of regular exercise — including reduced resting heart rate, improved vascular tone, and lower mean arterial pressure — may indirectly lower IOP through reductions in episcleral venous pressure, a known determinant of aqueous outflow resistance (Passo *et al.*, 1991; Qureshi, 1996). The near-identical mean blood pressure values in both groups in this study suggest that blood pressure is unlikely to fully explain the observed IOP difference; however, more chronic cardiovascular adaptations not captured by a single BP measurement may still play a role.

Thirdly, a growing body of evidence implicates systemic hormonal changes associated with chronic exercise — particularly changes in cortisol, growth hormone, and insulin sensitivity — as potential modulators of ciliary body function and aqueous production (Harris *et al.*, 1994). These hormonal pathways represent an area where further mechanistic research is warranted.

The mean difference of 1.91 mmHg observed in this study is clinically meaningful. The Ocular Hypertension Treatment Study (OHTS; Kass *et al.*, 2002) and the Early Manifest Glaucoma Trial (EMGT; Heijl *et al.*, 2002) both demonstrated that IOP reductions in the range of 1–3 mmHg are

associated with measurable reductions in glaucoma conversion and progression rates. While this study did not assess glaucoma risk per se, the observed IOP difference between lifestyle groups carries public health relevance.

5.1.2 Absence of Dose-Response Within Gym-Goers

The non-significant correlation between exercise frequency and IOP among gym-goers (Pearson $r = .139$, $p = .288$) is also consistent with some existing literature. Several studies have reported that while the transition from sedentary behavior to regular exercise is associated with meaningful IOP reduction, incremental increases in exercise frequency beyond a threshold do not yield proportionally greater IOP reductions (Williams *et al.*, 2013; Vera *et al.*, 2021). This suggests a threshold or saturation effect — once a participant attains regular gym participation, increasing the dose of exercise beyond that baseline may offer diminishing IOP returns.

It is also important to note that exercise frequency as measured in this study was an ordinal categorical variable reflecting the number of sessions per week, which does not capture exercise intensity, type, duration per session, or cumulative weekly load — all of which may be more sensitive predictors of chronic IOP changes than frequency alone. Future studies employing objective exercise monitoring (e.g., accelerometry, MET-based load calculations) would be better positioned to characterize a dose-response relationship if one exists.

5.1.3 Bilateral IOP Symmetry

Both groups showed high bilateral IOP symmetry — mean right and left eye IOP values were within 0.12 mmHg of each other in the gym-goer group, and within 0.15 mmHg in the non-gym-goer group. This is consistent with the expected physiological pattern of bilateral IOP concordance

in healthy adults without monocular disease (Dielemans *et al.*, 1994) and supports the validity of using IOP_Avg as the primary outcome variable rather than analyzing each eye independently.

5.1.4 Limitations

The following limitations are acknowledged and should be considered when interpreting the findings of this study:

- Cross-sectional design: The study's cross-sectional nature precludes causal inference. The observed association between gym participation and lower IOP does not establish that exercise caused the lower IOP — reverse causality (i.e., individuals with naturally lower IOP being more inclined to exercise) cannot be excluded.
- Self-reported exercise frequency: Exercise frequency was captured as a self-reported ordinal bracket, limiting the precision of the dose-response analysis and precluding the use of continuous, scale-level correlation techniques in their strictest sense.
- Sample size and balance: The study achieved the minimum target of 120 participants (60 per group), which was adequate for detecting the observed effect size. However, a larger sample would increase statistical power and allow for more robust subgroup analyses stratified by gender, age, or exercise type.
- Gender imbalance between groups: The gym-goer group was predominantly male (63.3%) while the non-gym-goer group was predominantly female (63.3%). Since gender has been reported as a potential influence on IOP in some studies, this imbalance may represent a confounding factor that future studies should control for through stratified sampling or covariate adjustment.

- Single IOP measurement: IOP was measured on a single occasion per participant. Diurnal variation in IOP is well-documented, and a single-point measurement may not fully capture a participant's true resting IOP. Repeated measurements at different times of day or on different days would improve accuracy.
- Time since last exercise not formally incorporated: While time since last exercise was collected as a variable, it was not formally incorporated into the main analysis. The known acute IOP-lowering effect of a single exercise session means that participants who exercised shortly before measurement may have artificially lower IOP readings, potentially exaggerating the group difference.

5.1.5 Contribution

This study makes several contributions to the body of knowledge on exercise, intraocular pressure, and ocular public health in Nigeria and sub-Saharan Africa:

- 5 It provides the first locally generated quantitative comparison of resting IOP between gym-going and non-gym-going adults specifically within Ilorin metropolis, addressing a gap in the literature on exercise-IOP relationships in the Nigerian context.
- 6 The use of bilateral IOP measurements (right eye, left eye, and average) with confirmed normality testing and Welch-adjusted inferential statistics adds methodological rigor to a literature where many existing Nigerian studies have relied on single-eye measurements or have not formally reported normality assessment.
- 7 The inclusion of effect size estimation (Cohen's $d = 0.68$) alongside the p-value enriches the clinical interpretability of the findings and enables future meta-analyses to incorporate this study's data quantitatively.

- 8 The findings support a public health narrative that regular gym participation may serve as a modifiable lifestyle factor with measurable ocular health benefits, providing a foundation for the integration of physical activity counselling into optometric practice in low- and middle-income settings.

5.2 Conclusion

This study provides empirical evidence from a Nigerian urban population that regular gym participation is associated with significantly lower resting intraocular pressure compared to a sedentary or non-gym lifestyle. The difference of approximately 1.91 mmHg between groups, while modest in absolute terms, reaches both statistical significance ($p < .001$) and clinical relevance, particularly in the context of glaucoma risk management where IOP reductions of comparable magnitude have been shown to slow disease progression.

The absence of a significant within-group dose-response relationship between exercise frequency and IOP suggests that the IOP-lowering benefit of gym participation may be realised through the distinction between being a regular exerciser versus not, rather than through a linear gradient of increasing exercise dose. This finding aligns with a threshold model of exercise-induced IOP modulation and underscores the importance of promoting any level of regular structured physical activity as a potential ocular health intervention.

Both hypotheses were formally tested, and the results were clear and consistent: H_{01} was rejected (significant between-group IOP difference), and H_{02} was retained (no significant exercise frequency–IOP correlation within gym-goers). Together, these findings contribute to the growing body of literature advocating for the role of lifestyle modification — and specifically habitual

physical exercise — in the non-pharmacological management of IOP and, by extension, in the primary prevention of glaucomatous optic neuropathy in susceptible populations.

In the specific context of Ilorin metropolis and the Nigerian healthcare landscape — where access to glaucoma medications and surgical interventions may be limited by cost and availability — the promotion of regular physical exercise as a low-cost, scalable IOP-management strategy holds substantial public health promise. Optometrists and primary eye care practitioners in this setting are well-positioned to counsel patients on the potential ocular health benefits of regular gym participation and structured exercise as part of holistic eye care delivery.

5.3 Recommendations

Based on the findings and conclusions of this study, the following recommendations are made:

8.3.1 Recommendations for Clinical and Public Health Practice

- Eye care practitioners — particularly optometrists — in Ilorin and across Nigeria should routinely incorporate physical activity counselling into their clinical workflow, advising patients (especially those at risk for elevated IOP or glaucoma) on the potential IOP-lowering benefits of regular structured exercise.
- Public health campaigns targeting modifiable risk factors for glaucoma in Nigeria should explicitly include the promotion of regular physical activity alongside well-established factors such as blood pressure control and dietary management.

- Gym facilities and fitness centers in Ilorin should be considered as community-based platforms for opportunistic eye health screening and IOP measurement, given the demonstrated association between gym participation and lower IOP observed in this study.
- Individuals identified as having borderline or elevated IOP during routine eye examinations should be actively encouraged to adopt a structured exercise programme, with follow-up IOP monitoring to evaluate the impact of lifestyle modification on intraocular pressure.

8.3.2 Recommendations for Future Research

- Future studies should employ objective physical activity monitoring instruments (accelerometers, fitness trackers, or MET-based load quantification) rather than self-reported ordinal frequency categories, to better characterise the quantitative exercise–IOP dose-response relationship.
- Longitudinal designs are warranted to determine the causal directionality of the observed association — specifically, whether initiating a gym exercise programme in previously sedentary individuals produces a measurable reduction in resting IOP over time.
- A randomized controlled trial (RCT) design, assigning sedentary participants with elevated IOP to a supervised exercise programme versus a control condition, would provide the strongest level of evidence for the causal role of exercise in IOP reduction.
- Future research should include a broader range of exercise types (aerobic versus resistance versus mind-body exercise such as yoga) to clarify which modalities confer the greatest chronic IOP-lowering benefits, given existing evidence of differential acute IOP responses to different exercise types.

- Studies with larger samples, greater geographic diversity across Nigerian cities, and gender-stratified analyses are needed to determine the generalizability of the current findings to broader Nigerian and West African populations.
- The potential confounding effects of diet, sleep quality, psychological stress, and hormonal factors on the exercise–IOP relationship should be systematically investigated and controlled for in future multivariate study designs.
- Mechanistic studies examining aqueous humor dynamics, trabecular outflow facility, and episcleral venous pressure in exercising versus non-exercising populations would help elucidate the biological pathways underlying the IOP differences documented in this and similar observational studies.

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APPENDIX

UNIVERSITY OF ILORIN

FACULTY OF LIFE SCIENCES

DEPARTMENT OF OPTOMETRY AND VISION SCIENCE

OPT698: RESEARCH PROJECT

Student's Name: Ofikwu Matthew Iduh

Supervisor's Name: Dr. H.E. Agbabiaka

*Project Topic: **COMPARATIVE ASSESSMENT OF RESTING INTRAOCULAR PRESSURE IN REGULAR GYM-GOERS AND NON-GYM-GOERS IN ILORIN METROPOLIS, KWARA STATE.***

QUESTIONNAIRE / DATA COLLECTION SHEET

Comparative Assessment of Resting Intraocular Pressure in Regular Gym Goers and Non-Gym Goers in Ilorin Metropolis, Kwara State

Participant ID: _____

Date of Examination: _____

Location (Gym/Clinic): _____

Time of Test: _____

SECTION A: DEMOGRAPHIC INFORMATION

1. Age: _____ years

2. Gender: ☐ Male ☐ Female

3. Occupation: _____

4. Workplace/Department (if applicable): _____

5. Residence (Area in Ilorin): _____

SECTION B: PHYSICAL ACTIVITY / GYM PARTICIPATION

6. Do you regularly attend a gym? ☐ Yes ☐ No

7. If yes, how long have you been attending the gym?

☐ Less than 6 months ☐ 6 months – 1 year ☐ 1 – 3 years ☐ More than 3 years

8. How often do you exercise per week?

☐ 1–2 times ☐ 3–4 times ☐ 5–6 times ☐ Daily

9. Average duration of exercise per session:

☐ Less than 30 minutes ☐ 30–60 minutes ☐ More than 60 minutes

10. Type of exercise commonly performed:

☐ Weightlifting / resistance training ☐ Cardio exercise ☐ Mixed training ☐ Others:

11. Time since last exercise session:

☐ Less than 12 hours ☐ 12–24 hours ☐ More than 24 hours

SECTION C: MEDICAL HISTORY

12. Do you have a history of the following?

Hypertension: ☐ Yes ☐ No

Diabetes: ☐ Yes ☐ No

Glaucoma: ☐ Yes ☐ No

Eye injury: ☐ Yes ☐ No

13. Are you currently using any medications? ☐ Yes ☐ No

If yes, specify: _____

14. Are you currently using any eye drops? ☐ Yes ☐ No

If yes, specify: _____

15. Family history of glaucoma? ☐ Yes ☐ No ☐ Not sure

SECTION D: LIFESTYLE FACTORS

16. Do you smoke? ☐ Yes ☐ No

17. Do you consume alcohol? ☐ Yes ☐ No

18. Caffeine intake (coffee/energy drinks):

☐ None ☐ Occasionally ☐ Regularly

SECTION E: CLINICAL MEASUREMENTS (TO BE FILLED BY RESEARCHER)

Blood Pressure:

Systolic BP: _____ mmHg

Diastolic BP: _____ mmHg

Pulse Rate: _____ bpm

Height: _____ cm

Weight: _____ kg

BMI: _____ kg/m²

SECTION F: OCULAR MEASUREMENTS

Right Eye IOP: _____ mmHg

Left Eye IOP: _____ mmHg

Instrument Used: _____

Examiner's Name: _____

CONSENT

I voluntarily agree to participate in this study and understand that the information provided will be used strictly for research purposes.

Participant Signature: _____

Researcher Signature: _____